Xucong **Hu**

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Education

Zhejiang University (ZJU) *Hangzhou, China*
M.S. in Psychology *Sep. 2024 - Jun. 2027 (Expected)*

- **GPA:** 4.0/4.0 (1/60)
- **Advisor:** Jifan Zhou, Mowei Shen
- **Award:**
  - National Scholarship (2024-2025, top 1 student in college per year)
  - First-Class Academic Scholarship (2024-2025, top 40% student in college per year)

Southwest University (SWU) *Chongqing, China*
B.S. in Psychology *Sep. 2020 - Jun. 2024*

- **GPA:** 4.13/5.0 (2/103)
- **Award:**
  - First-Class Academic Scholarship (2021–2023, top 3 students in college per year)
  - Excellent Undergraduate Thesis Award (2024, Top 10% among graduates in college)
  - Challenge Cup Bronze Award (2022, Top 1% among college projects, Provincial-level award)
  - Provincial Undergraduate Training Program for Innovation and Entrepreneurship, (2023, awarded to top 1% of undergraduates)

Publication

Published

- **Hu, X.**, Xu, H., Chen, H., Shen, M., & Zhou, J. (2025). Good to see you R2-D2: Inducing spontaneous perspective-taking towards non-human agents through human-like gaze and reach. *Cognition, 259*, 106101.
- Zhao, Y., **Hu, X.**, Zhou, J., Shen, M., & Xu, H. (2025). Enhancement of joint flanker effect in intergroup competition. *PsyCh Journal, 14*(1), 94–102.
- **Hu, X.**, & Tong, S. (2023). Effects of robot animacy and emotional expressions on perspective-taking abilities: a comparative study across age groups. *Behavioral Sciences, 13*(9), 728.
- Zhu, J., Yang, Z., Ma, R., Yin, L., & **Hu, X**. (2022). Face beauty or soul beauty? The influence of facial attractiveness and moral judgment on pain empathy. *Frontiers in Psychology, 13*, 990637.
- **Hu, X.**, Zhou, J., & Shen, M. (2024, Dec). Spontaneous perspective-taking toward multi-view robots: Evidence supporting the social agent model. *Annual Conference on Industrial Psychology and New Productivity*. **(Oral Report)**.
- **Hu, X.** (2022, May). The uncanny valley effect and induced compensatory consumption behavior in language conditions. *Lecture Notes in Education Psychology and Public Media, 4*, 426–436. **(Oral Report)**

Under Review

- **Hu, X.**, Xu, H., Chen, H., Shen, M., & Zhou, J. (2025). Seeing through Janus' Eyes: How Humans Spontaneously Adopt Multiple Perspectives of Robot Avatars. (under review at *International Journal of Social Robotics)*
- **Hu, X.**, Xu, E., Xu, H., Shen, M., & Zhou, J. (2025). Causal Relationship Between Robot Perception and Behavior: A User-Centered Explainability Approach. (under review at *International Journal of Human-Computer Interaction)*
- **Hu, X.**, Zheng, Y., Hu, Q., Chen, H., Shen, M., & Zhou, J. (2025). I'll Believe It Unless It's Too Absurd: Spontaneous Visual Perspective-Taking as Prior-Based Heuristic Inference. (under review at *Cognition)*
- **Hu, X.**, Hu, Q., Yu, T., Shen, M., & Zhou, J. (2025). Do Robots Need a Head and Legs? How Appearance Features Predict Robots' Perceived Social Interaction Potential (under review at *ACM/IEEE International Conference on Human Robot Interaction (HRI))*

Research Experience

Project Leader | Social Cognition underlying Human-Robot Interaction *Hangzhou, China*
Supervised by Prof. Jifan Zhou & Prof.Mowei Shen, Zhejiang University *Jun. 2023 - Current*

- Designed and implemented online cognitive psychology experiments using JavaScript and HTML, collecting behavioral data (reaction times, accuracy) from 3,000+ participants.
- Conducted statistical analyses in R (ANOVA, t-test, logistic regression, GLMM) on behavioral datasets.
- Performed exploratory and confirmatory factor analyses and reliability/validity testing on questionnaire data from 750 participants using R.
- Built predictive models of user ratings on robots with Python (scikit-learn, XGBoost), applying SHAP for interpretable machine learning.
- Created publication-quality visualizations using R (ggplot2) and Python (matplotlib).
- Collected eye-tracking data with Tobii, integrated participants' verbal reports with video-based behavioral coding, and applied Hidden Markov Models (HMMs) in PyMC for cognitive process modeling (ongoing).
- Produced 6+ peer-reviewed manuscripts in psychology, robotics, and computer science, with publications in leading journals such as Cognition.

Remote Research Assistant | LLM–Human Differences in Theory of Mind *Hong Kong, China*

SUPERVISED BY PROF. JIANQIAO ZHU, THE UNIVERSITY OF HONG KONG *Oct. 2025 – Present*

- Designed a dynamic grid-world Theory-of-Mind task to benchmark LLMs through API-based evaluations.
- Implemented Bayesian belief updating to compare inference trajectories between large language models (LLMs) and human data, leveraging MCMC-based power sampling to reveal fundamental differences in their underlying computational mechanisms.

Research Assistant | Cognitive Load in Command Decision-Making *Beijing, China*

SUPERVISED BY PROF. JINGYU ZHANG, CHINESE ACADEMY OF SCIENCES, INSTITUTE OF PSYCHOLOGY *Apr. 2023 - Aug. 2023*

- Simulated multitasking decision-making tasks using the strategy game Command: Modern Operations.
- Conducted semi-structured expert interviews with professional commanders.
- Collected eye-tracking data with Tobii and physiological measures (heart rate, skin conductance) with BIOPAC.
- Analyzed behavioral data with R (ANOVA, t-test) and created visualizations with ggplot2.

Remote Research Assistant | Perspective-Taking toward Robots in Older Adults *Beijing, China*

SUPERVISED BY DR. SONG TONG, TSINGHUA UNIVERSITY *Sep. 2022 - Aug. 2023*

- Created human-like face stimuli with varying anthropomorphism using FantaMorph.
- Programmed experimental tasks with PsychoPy (Python).
- Recruited and assisted older adult participants from community settings in completing tests and questionnaires.
- Analyzed behavioral data in SPSS using ANOVA and t-tests.

Skill

CODING

- **Statistics:** R, Python (statsmodels, scipy, PyMC), SPSS, Mplus, Bayesian Modeling, Factor Analysis, Structural Equation Modeling
- **Machine Learning:** Python (scikit-learn, XGBoost, PyTorch, SHAP), Reinforcement Learning (Q-learning)
- **Frontend:** HTML, CSS, JavaScript, React, NEXT.js
- **Backend:** Node.js
- **Software Design:** Photoshop, Figma, Unity, Blender

LANGUAGE

- **English:** GRE (159+169+4.0), IELTS (7.0)
- **Chinese:** Native

Teaching

The Advanced Psychological Measurement *Hangzhou, China*

TEACHING ASSISTANT, DEPARTMENT OF PSYCHOLOGY AND BEHAVIORAL SCIENCES, ZHEJIANG UNIVERSITY *Apr. 2025 - Jun. 2025*

- Delivered presentations introducing commonly used experimental paradigms in psychology to enhance students' research design skills.
- Explained the mathematical foundations of advanced psychological methods, including cognitive modeling approaches, during class and office hours.

Reviewer

- **Behavioral Sciences** (SSCI Q2)

References

Jifan Zhou

PROFESSOR

- Department of Psychology and Behavioral Sciences, Zhejiang University
- jifanzhou@zju.edu.cn

Mowei Shen

PROFESSOR

- Department of Psychology and Behavioral Sciences, Zhejiang University
- mwshen@zju.edu.cn

Haokui Xu

ASSISTANT PROFESSOR

- Institute of Applied Psychology, Zhejiang University of Technology
- haokuixu@zjut.edu.cn